

CMU 10-703: Deep Reinforcement Learning

Lecture 7: Advanced Policy Gradient Methods

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Hybrid Policy Optimization

The target is an explicit policy that can improve from data collected by older policies.



Off-policy data

Reuse samples from old policies.

Replay

Store and revisit past transitions.

Explicit policy

Train π_θ rather than an argmax rule.

1.

Performance Difference Lemma

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Performance Difference Lemma

Lemma. For any policies $\pi(a \mid s)$ and $\mu(a \mid s)$,

$$J(\pi) - J(\mu) = \mathbb{E}_{\tau \sim P_{\pi}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\mu}(s_t, a_t) \right].$$

Expected return

$$J(\pi) = \mathbb{E}_{\tau \sim P_{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

Reference-policy advantage

$$A^{\mu}(s, a) = Q^{\mu}(s, a) - V^{\mu}(s)$$

Visit states under π and measure its actions relative to μ .

Equivalent Improvement Objective

Reference policy. Fix μ while optimizing the candidate policy π .

$$\arg \max_{\pi} J(\pi) = \arg \max_{\pi} (J(\pi) - J(\mu)).$$



Policy iteration can optimize improvement over a fixed reference.

Proof Setup

Return from the initial state

$$J(\pi) = \mathbb{E}_{s_0 \sim P(s_0)}[V^\pi(s_0)].$$

Extend the expectation

$$J(\pi) = \mathbb{E}_{\tau \sim P_\mu(\tau)}[V^\pi(s_0)].$$

$V^\pi(s_0)$ depends on the sampled trajectory only through s_0 .

Telescoping Value Identity

$$V^\pi(s_0) = \sum_{t=0}^{\infty} \gamma^t V^\pi(s_t) - \sum_{t=1}^{\infty} \gamma^t V^\pi(s_t)$$

The two infinite sums cancel except for the initial value.

Telescoping Value Identity

$$\begin{aligned} V^\pi(s_0) &= \sum_{t=0}^{\infty} \gamma^t V^\pi(s_t) - \sum_{t=1}^{\infty} \gamma^t V^\pi(s_t) \\ &= \sum_{t=0}^{\infty} \gamma^t V^\pi(s_t) - \sum_{t=0}^{\infty} \gamma^{t+1} V^\pi(s_{t+1}) \end{aligned}$$

Reindex the second sum to align time steps.

Telescoping Value Identity

$$\begin{aligned} V^\pi(s_0) &= \sum_{t=0}^{\infty} \gamma^t V^\pi(s_t) - \sum_{t=1}^{\infty} \gamma^t V^\pi(s_t) \\ &= \sum_{t=0}^{\infty} \gamma^t V^\pi(s_t) - \sum_{t=0}^{\infty} \gamma^{t+1} V^\pi(s_{t+1}) \\ &= \sum_{t=0}^{\infty} \gamma^t (V^\pi(s_t) - \gamma V^\pi(s_{t+1})). \end{aligned}$$

The initial value becomes a discounted sum of one-step differences.

One-Step Advantage Identity

$$Q^{\pi}(s_t, a_t) = r(s_t, a_t) + \gamma V^{\pi}(s_{t+1})$$

Expand the action value through one transition.

One-Step Advantage Identity

$$Q^{\pi}(s_t, a_t) = r(s_t, a_t) + \gamma V^{\pi}(s_{t+1})$$

$$V^{\pi}(s_t) - \gamma V^{\pi}(s_{t+1}) = V^{\pi}(s_t) - Q^{\pi}(s_t, a_t) + r(s_t, a_t)$$

Substitute for the successor value.

One-Step Advantage Identity

$$Q^\pi(s_t, a_t) = r(s_t, a_t) + \gamma V^\pi(s_{t+1})$$

$$\begin{aligned} V^\pi(s_t) - \gamma V^\pi(s_{t+1}) &= V^\pi(s_t) - Q^\pi(s_t, a_t) + r(s_t, a_t) \\ &= -A^\pi(s_t, a_t) + r(s_t, a_t). \end{aligned}$$

Recognize the advantage $A^\pi = Q^\pi - V^\pi$.

One-Step Advantage Identity

$$Q^\pi(s_t, a_t) = r(s_t, a_t) + \gamma V^\pi(s_{t+1})$$

$$\begin{aligned} V^\pi(s_t) - \gamma V^\pi(s_{t+1}) &= V^\pi(s_t) - Q^\pi(s_t, a_t) + r(s_t, a_t) \\ &= -A^\pi(s_t, a_t) + r(s_t, a_t). \end{aligned}$$

$$V^\pi(s_0) = - \sum_{t=0}^{\infty} \gamma^t A^\pi(s_t, a_t) + \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t).$$

Insert the identity into the telescoping sum.

Completion of the Proof

$$J(\pi) = -\mathbb{E}_{\tau \sim P_\mu} \left[\sum_{t=0}^{\infty} \gamma^t A^\pi(s_t, a_t) \right] + J(\mu)$$

Under P_μ , the reward sum has expectation $J(\mu)$.

Completion of the Proof

$$J(\pi) = -\mathbb{E}_{\tau \sim P_\mu} \left[\sum_{t=0}^{\infty} \gamma^t A^\pi(s_t, a_t) \right] + J(\mu)$$

$$J(\mu) - J(\pi) = \mathbb{E}_{\tau \sim P_\mu} \left[\sum_{t=0}^{\infty} \gamma^t A^\pi(s_t, a_t) \right].$$

Rearrange to express the improvement of μ over π .

Completion of the Proof

$$J(\pi) = -\mathbb{E}_{\tau \sim P_\mu} \left[\sum_{t=0}^{\infty} \gamma^t A^\pi(s_t, a_t) \right] + J(\mu)$$

$$J(\mu) - J(\pi) = \mathbb{E}_{\tau \sim P_\mu} \left[\sum_{t=0}^{\infty} \gamma^t A^\pi(s_t, a_t) \right].$$

$$J(\pi) - J(\mu) = \mathbb{E}_{\tau \sim P_\pi} \left[\sum_{t=0}^{\infty} \gamma^t A^\mu(s_t, a_t) \right].$$

Swap π and μ to obtain the stated lemma. $\square$

Uses of the Lemma

Policy comparison

Measure improvement relative to a reference policy.

Policy iteration

Repeatedly update a policy to improve on its predecessor.

Advantage centering

Remove state-dependent baselines and compare actions.

Off-policy bridge

Separate the policy that supplies data from the policy optimized.

$$\text{positive expected } A^\mu \implies J(\pi) > J(\mu).$$

2.

Discounted State Distributions

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Trajectory and State-Distribution Expectations

Discounted state distribution. d^π is the normalized discounted distribution of states visited by policy π .

$$\mathbb{E}_{\tau \sim P_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \right] = \frac{1}{1 - \gamma} \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi(\cdot|s)}} [f(s, a)].$$

Left side

A cumulative discounted sum along trajectories.

Right side

A normalized expectation over states and actions.

Discounted State-Distribution Recursion

$$d^\pi(s') = (1 - \gamma)p_0(s') + \gamma \sum_{s,a} d^\pi(s) \pi(a | s) P(s' | s, a).$$



The recursion propagates discounted probability mass forward.

State-Distribution Form of the Lemma

$$J(\pi) - J(\mu) = \frac{1}{1 - \gamma} \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi(\cdot|s)}} [A^\mu(s, a)].$$



The candidate policy controls both its state distribution and actions.

3.

Hybrid Objectives

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Two Policy-Distribution Pairings

States from π , advantages from μ

$$\mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [A^\mu(s, a)]$$

Estimate A^μ from rollouts or a critic. The state expectation still requires the current policy.

PPO and AWR

States from μ , advantages from π

$$\mathbb{E}_{\substack{s \sim d^\mu \\ a \sim \pi}} [A^\pi(s, a)]$$

Reuse replay data, but learn Q^π and A^π off-policy.

SAC and TD3

Both pairings decouple data collection from policy optimization.

Behavior-Data Approximation



$$d^\pi \approx d^\mu.$$

Benefit

Evaluate the objective on states already observed under μ .

Requirement

Keep the candidate policy close enough to the behavior policy.

Exact State-Reweighted Objective

Importance sampling. For state functions f , reweight samples from d^μ by the state-density ratio d^π/d^μ .

$$J(\pi) - J(\mu) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\mu} \left[\frac{d^\pi(s)}{d^\mu(s)} \mathbb{E}_{a \sim \pi(\cdot|s)} [A^\mu(s, a)] \right].$$

The state-density ratio makes the behavior-data expression exact.

Approximate Behavior-State Objective

Drop the state-density ratio

$$J(\pi) - J(\mu) \approx \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\mu} \left[\mathbb{E}_{a \sim \pi(\cdot|s)} [A^\mu(s, a)] \right].$$

Tractable dependence

π_θ appears only through its action distribution at states visited by μ .

Learned estimate

Replace A^μ by an estimator such as A_ϕ^μ .

The approximation is credible only while d^π remains near d^μ .

State Importance-Ratio Tradeoff

Keep d^π / d^μ

exact objective



high variance and
unknown transitions

Drop d^π / d^μ

tractable objective



requires nearby
state distributions

Practical methods replace state reweighting with controlled policy updates.

4.

Divergence Regularization

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Divergence-Regularized Objective

$$\max_{\pi} \mathbb{E}_{s \sim d^{\mu}} \left[\mathbb{E}_{a \sim \pi(\cdot | s)} [A^{\mu}(s, a)] - \alpha D(\pi(\cdot | s), \mu(\cdot | s)) \right].$$

Improvement term

Shift probability toward actions with positive A^{μ} .

Proximity term

Penalize changes that invalidate $d^{\pi} \approx d^{\mu}$.

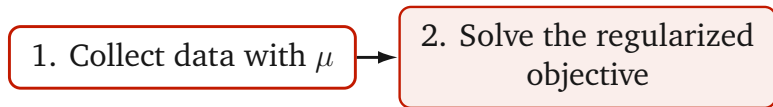
α balances policy improvement against distribution shift.

Regularized Policy-Iteration Loop

1. Collect data with μ

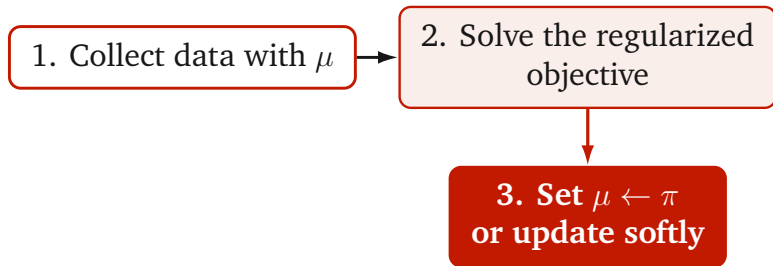
The behavior policy is initialized arbitrarily and supplies data.

Regularized Policy-Iteration Loop



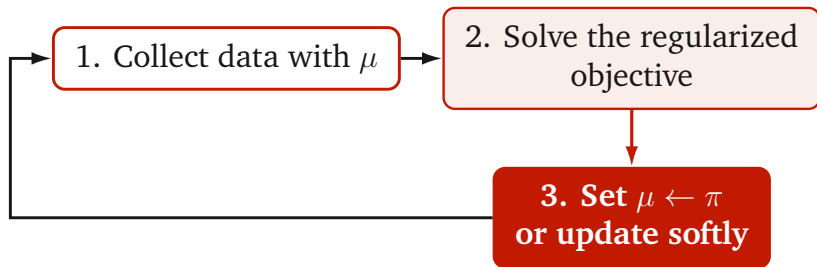
The optimization step produces a nearby improved policy.

Regularized Policy-Iteration Loop



The behavior policy follows the improved policy.

Regularized Policy-Iteration Loop



repeat for $i = 0, \dots, T$.

Data collection and optimization advance together.

Total Variation Distance

Total variation. At state s , TV measures the difference between the action probabilities assigned by two policies.

$$D_{\text{TV}}(\pi(\cdot | s), \mu(\cdot | s)) = \frac{1}{2} \int_{\mathcal{A}} |\pi(a | s) - \mu(a | s)| da.$$

Theoretical role

Small per-state TV controls the shift from d^μ to d^π .

Source: Achiam et al., “Constrained Policy Optimization,” ICML 2017.

Optimization cost

TV is difficult to optimize in high-dimensional action spaces.

Policy Closeness and State Distributions



Conservative action-distribution changes limit the resulting shift in discounted state visitation.

$$D_{\text{TV}}(\pi(\cdot | s), \mu(\cdot | s)) \text{ small} \implies d^\pi \text{ near } d^\mu.$$

Source: Achiam et al., "Constrained Policy Optimization," ICML 2017.

KL Divergence and Pinsker's Inequality

Kullback–Leibler divergence

$$D_{\text{KL}}(p \parallel q) = \int p(a \mid s) \log \frac{p(a \mid s)}{q(a \mid s)} da.$$

Pinsker's inequality

$$D_{\text{TV}}(p, q) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(p \parallel q)}, \quad D_{\text{TV}}(p, q) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(q \parallel p)}.$$

Controlling KL in either direction controls symmetric TV distance.

Two KL Directions

Forward KL

$$D_{\text{KL}}(\pi \parallel \mu)$$

Advantage-Weighted Regression

Closed-form nonparametric optimizer.

Source: Peng et al., AWR, 2019.

Reverse KL

$$D_{\text{KL}}(\mu \parallel \pi)$$

TRPO and PPO

Discourages negligible probability on actions explored by μ .

Source: Schulman et al., TRPO, 2015; PPO, 2017.

5.

Advantage-Weighted Regression

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AWR Objective

$$\max_{\pi} \mathbb{E}_{s \sim d^{\mu}} \left[\mathbb{E}_{a \sim \pi(\cdot | s)} [A^{\mu}(s, a)] - \alpha D_{\text{KL}}(\pi(\cdot | s) \parallel \mu(\cdot | s)) \right] .$$

Reward for change

Favor actions with larger reference-policy advantage.

Cost of change

Stay close to the behavior policy in forward KL.

Source: Peng et al., "Advantage-Weighted Regression," 2019.

KL-Regularized Closed Form

Generic problem. Optimize a distribution p relative to a reference distribution q .

$$\max_{p(x)} \{ \mathbb{E}_{x \sim p}[f(x)] - \alpha D_{\text{KL}}(p \parallel q) \}.$$

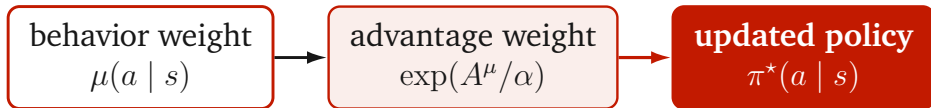
Functional optimum

$$p^*(x) \propto q(x) \exp \left(\frac{f(x)}{\alpha} \right).$$

Exponentiated scores reweight the reference distribution.

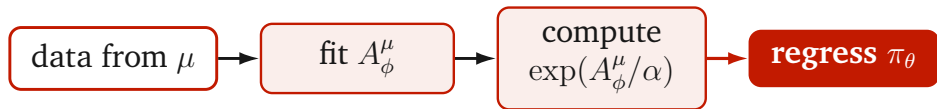
AWR Policy Update

$$\pi^*(a | s) \propto \mu(a | s) \exp \left(\frac{A^\mu(s, a)}{\alpha} \right).$$



Positive-advantage actions gain probability relative to μ .

Practical AWR Procedure



Supervised value fitting

Estimate the behavior-policy advantage from stored data.

Weighted policy fitting

Regress the policy toward actions weighted by estimated advantage.

AWR converts a regularized policy update into two supervised steps.

Advanced Policy Gradient Methods

Exact identity

The performance difference lemma expresses improvement through d^π , π , and A^μ .

Practical approximation

Substitute d^μ for d^π to learn from behavior data.

Distribution control

TV explains the required closeness; KL supplies a tractable surrogate.

AWR

Forward-KL regularization yields an exponentiated-advantage update.

$$\pi^*(a \mid s) \propto \mu(a \mid s) \exp \left(\frac{A^\mu(s, a)}{\alpha} \right).$$